

B. Christensen (2025), "Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi". Research Report protected by copyright laws.

Benjamin Christensen

Aarhus University

***Preprint (AFI Research report 2024/25). This paper is currently under submission. The content is protected by copyright law. No parts of the text may be cited without the author's permission***

# Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi

---

## 1. Introduction

Understanding the causal dynamics of social-ecological systems is critical for effectively addressing environmental and societal challenges. These systems, however, are inherently complex, being characterized by physical, biological, ecological, and human interactions that evolve over time, and adaptive behavior where changes within individual parts can lead to unpredictable system-wide outcomes. Since conventional linear models may fail to capture the feedback loops and emergent properties that characterize social-ecological systems, it is challenging to isolate and identify their causal properties (Schlüter et al. 2023; Banitz et al. 2022).

This study utilizes the software package “PyPhi” to quantify the complex causal dynamics articulated by an agent-based model designed to simulate the relationship between overfishing and hypoxia in marine ecosystems. The “integrated information” gauged within the Integrated Information Theory framework is of particular interest as it offers quantifiable insights into how information is processed and integrated across different parts of systems, reflecting their capacity for cohesion, resilience, and possible responses to external pressures. Such an analysis may provide novel perspectives on the interdependencies and potential vulnerabilities within social-ecological systems, and the approach thus not only enhances our understanding of the causal mechanisms underlying one type of complex system, but also of the available leverage points for interventions promoting sustainability and resilience.

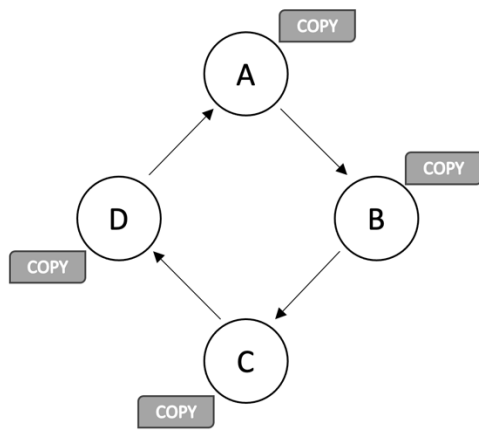
## 2. Integrated Information Theory

One of the main appeals of Integrated Information Theory (IIT) is that it uses information theoretic tools to identify system behavior that cannot be reduced to the behavior of underlying parts—that is, it formally describes what it is about a system that makes it “more than the sum of its parts” (Mediano et al. 2022, 2). More concretely, IIT’s approach is to describe causal interdependencies between subsystems within systems in terms of mutual

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

information (Tononi 2004; Tononi et al. 1996). When all subsystems exhibit behavior that they do not outside of their interactions within that system, IIT considers this to signify a causally cohesive system with its own properties over and above those characterizing its parts (Albantakis et al. 2023, 10-11).

The formal foundation of IIT consists in mapping transition probabilities between system states via Markov matrices. Systems are represented as Boolean networks with discrete time series dynamics, as illustrated in figure 1 (Mayner et al. 2018, 2-4).

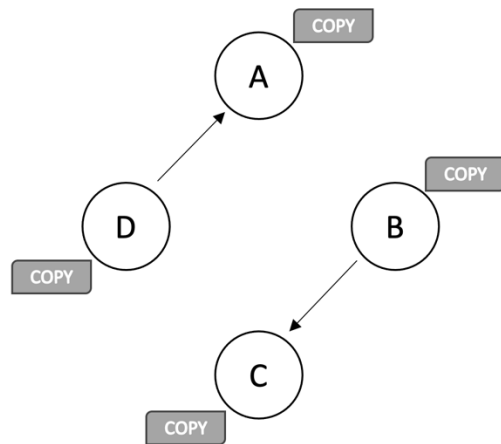


**Fig. 1**

A Boolean network consisting of four nodes: “A”, “B”, “C”, and “D”. Each node has a copy function, so that it will at  $t$  copy the state at  $t-1$  of the node outputting to it.

For such a system, A is observed to change its state  $S$  at  $t+1$  only if D has been active at  $t$ , B at  $t+2$  only if A has been active at  $t+1$ , C at  $t+3$  only if B has been active at  $t+2$ , and finally D at  $t+4$  only if C has been active at  $t+3$  (Oizumi et al. 2014, 7). The algorithms utilized by IIT capture the mutual information between units by quantifying the probability with which they affect and are affected by each other (Albantakis et al. 2023, 8-11). For the system in figure 1, for instance, the probability that A is active at  $t+1$  is 1.0 if D was active at  $t$ .

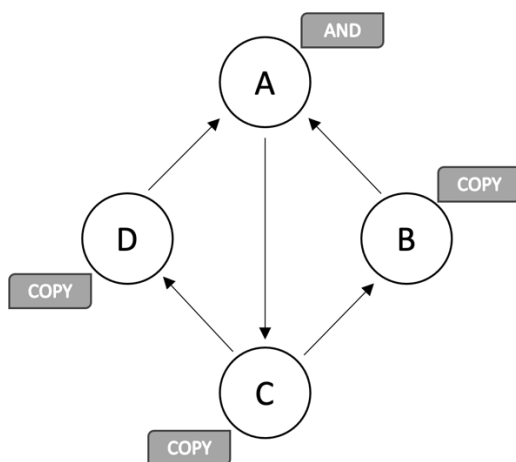
On the flip side, if no causal links can be established between its purported parts, this indicates according to IIT that we are in fact not dealing with one cohesive system, but rather several separate systems, as illustrated in figure 2.



**Fig. 2**

The Boolean system seen in figure 1 has here been effectively separated into two independent systems by blocking the outputs from A to B and C to D.

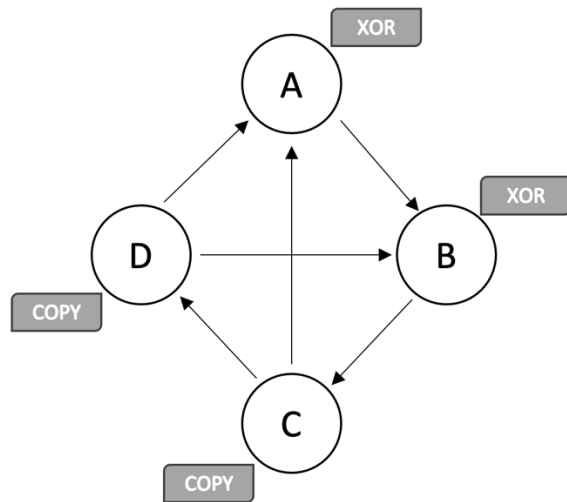
In more highly integrated systems, the states of subsystems are affected by multiple other subsystems at a time, such as in figure 3 below, where A is only active at  $t+1$  if D as well as B were active at  $t$ .



**Fig. 3**

A Boolean network consisting of four nodes like the networks seen in figures 1 and 2. Here, however, the behavior of one node is dependent on two other nodes, since A will activate at  $t$  only if both B and D were active at  $t_1$ .

It may also be that the causal relationships between subsystems are more indirect, such as in figure 4, where B responds at  $t+1$  to the activity of A at  $t$  only if A is not active in combination with D.



**Fig. 4**

A Boolean network with more complex interactions, where the behavior of B is dependent not only upon the independent activities of A and D, but also upon the activities of A relative to D and vice-versa, since B will only activate at t if *either A or D* was active at  $t-1$ .

Note that A constrains the state of B with probability 0.5, whereas the combined activity of A and D constrains B with probability 1.0. IIT considers as possible parts of a system not only individual units such as “A”, “B”, “C” and “D”, but also any composition of these that itself has causal properties not reducible to the causal properties of the individual units. Thus, “AD” is a part of the system in figure 4 since the effects of A and D individually on B do not sum to the effects of A and D relative to each other on B (Oizumi et al 2014, 8-9).

### 3. Integrated Information Theory and Complex Systems

IIT was designed to describe and measure correlations between the emergence of conscious states in brains and causal interdependencies between their various subsets (Tononi et al. 1996; 2004). While the precise relationship between integrated information and conscious states is sometimes construed in a manner that is still debated within the field of neuroscience<sup>1</sup>, the generalized application of the theory’s formal framework to complex systems does not intrinsically carry such implications. Thus, researchers specializing in IIT have successfully shown that the formalism can be generalized to study causal relationships and emergence within any type of physical system (Albantakis et al. 2019; Marshall et al.

---

<sup>1</sup> That is, they argue that “maximal integrated information” in any physical system is a necessary *and sufficient* condition for consciousness. See Albantakis et al. 2023 pages 5-9 for more details.

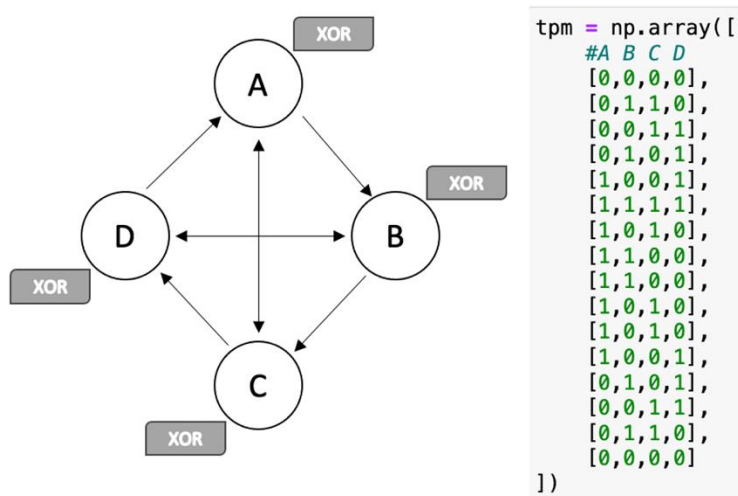
B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

2018), and consequently IIT formalism has been applied to issues as different as quantum entanglement (Albantakis et al. 2023), dynamical and information-processing complexity (Mediano et al. 2022), the cell cycle in fission yeast (Marshall et al. 2017), the dynamics of fish schooling (Niizato et al. 2020), the dynamics of group evolution (Fischer et al. 2020), capturing informational dynamics in social groups (Engel & Malone 2018) and the emergence of social entities relative to their parts (Christensen 2023).

## 4. PyPhi

### a. Markovian Dynamics

“PyPhi” is a Python library developed for the computational management of integrated information by William Mayner, William Marshall, Larissa Albantakis, Graham Findlay, Robert Marchman and Tononi (2018).<sup>2</sup> Based on the mathematical approach presented by Masafumi Oizumi, Albantakis, and Tononi in their 2014 (Mayner et al. 2018, 2),<sup>3</sup> the first step in employing the software is to define a Markov matrix representing the system to be analyzed (Mayner et al. 2018, 2-4), as shown in figure 5.



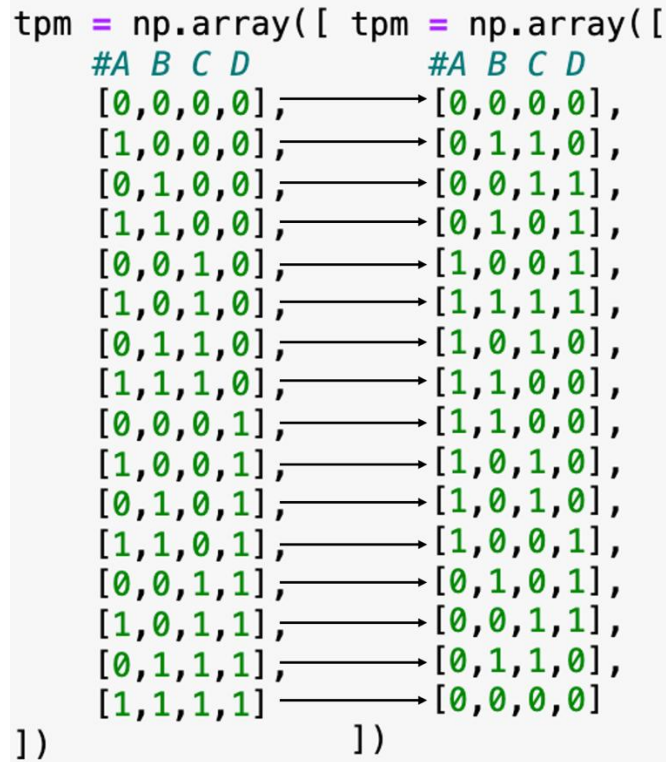
**Fig. 5**

To the left is seen a Boolean network consisting of four nodes. To the right is seen the Markov (transition probability) matrix representing that network, as described within PyPhi (Mayner et al. 2018, 4).

<sup>2</sup> The following is only a brief overview meant to introduce the PyPhi functionalities that are utilized within the present work. For the full scope of PyPhi, see the highly detailed documentation generously shared by its developers at <https://pyphi.readthedocs.io/en/latest/installation.html>.

<sup>3</sup> Likewise, the math behind IIT is not here delved into. For detailed renditions of this see Oizumi et al. 2014, Mayner et al. 2018, and Albantakis et al. 2023.

The Markov matrix defines all possible transitions between all possible system states (Mayner et al. 2018, 2-3), as shown in figure 6.

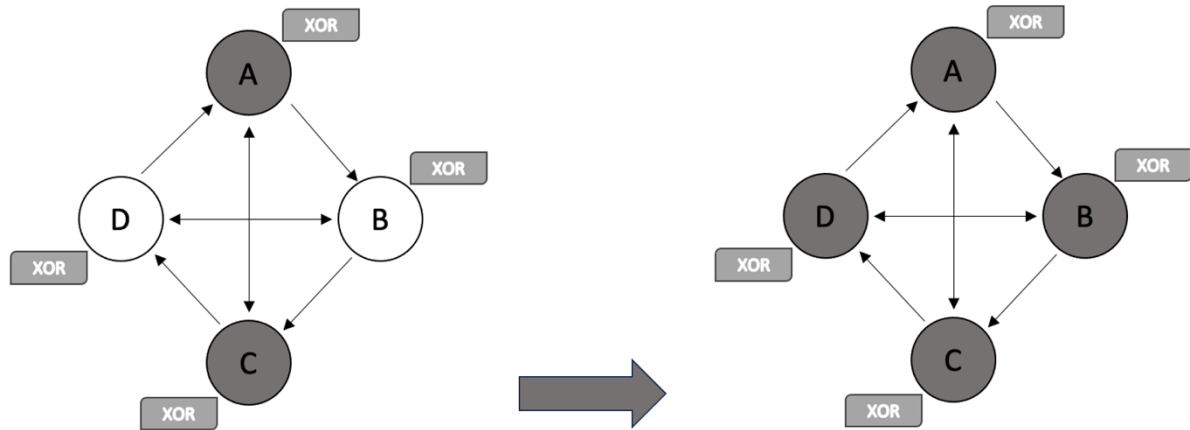


**Fig. 6**

An illustration of the states of the system seen in figure 5 that precede the ones represented by its Markov matrix.

PyPhi uses the Markov matrix passed to it to calculate the probability that one state of the system will transition into another (Mayner et al. 2018, 2-4). For instance, if in the system seen in figure 5 the nodes A and C are currently active while B and D remain inactive—conventionally represented by the notation “ABCD = 1010” (Oizumi et al. 2014, 5)—then there is a probability 1.0 that the next state will be ABCD = 1111 as shown in figure 7.

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

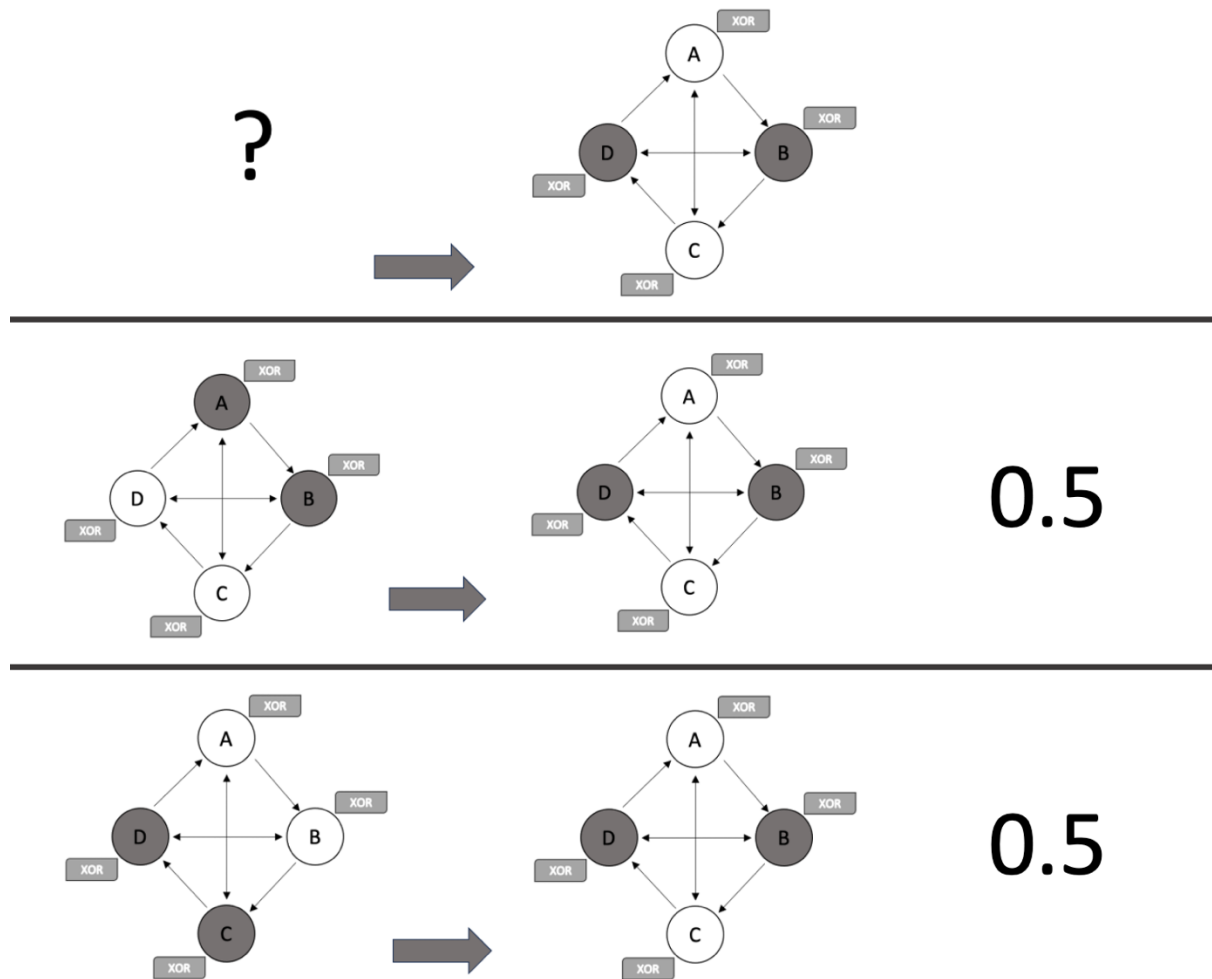


**Fig. 7**

An illustration of the only possible transition for the system seen in figure 5 above from the state  $ABCD = 1010$ . The effect of the system being in the state  $ABCD = 1010$  at  $t$  is that it is in the state  $ABCD = 1111$  at  $t+1$ .

Transition probabilities are calculated both from the previous state of the system to the current one and from the current state to the next; in other words, for any possible state probabilities are calculated for all the possible causes and effects of that state found within the system (Mayner et al. 2018, 7-8). In many cases, more than one state may precede or follow from the current state being considered, as shown in figure 8 below where either the state  $ABCD = 1100$  or the state  $ABCD = 0011$  could have preceded the current state  $ABCD = 0101$ .

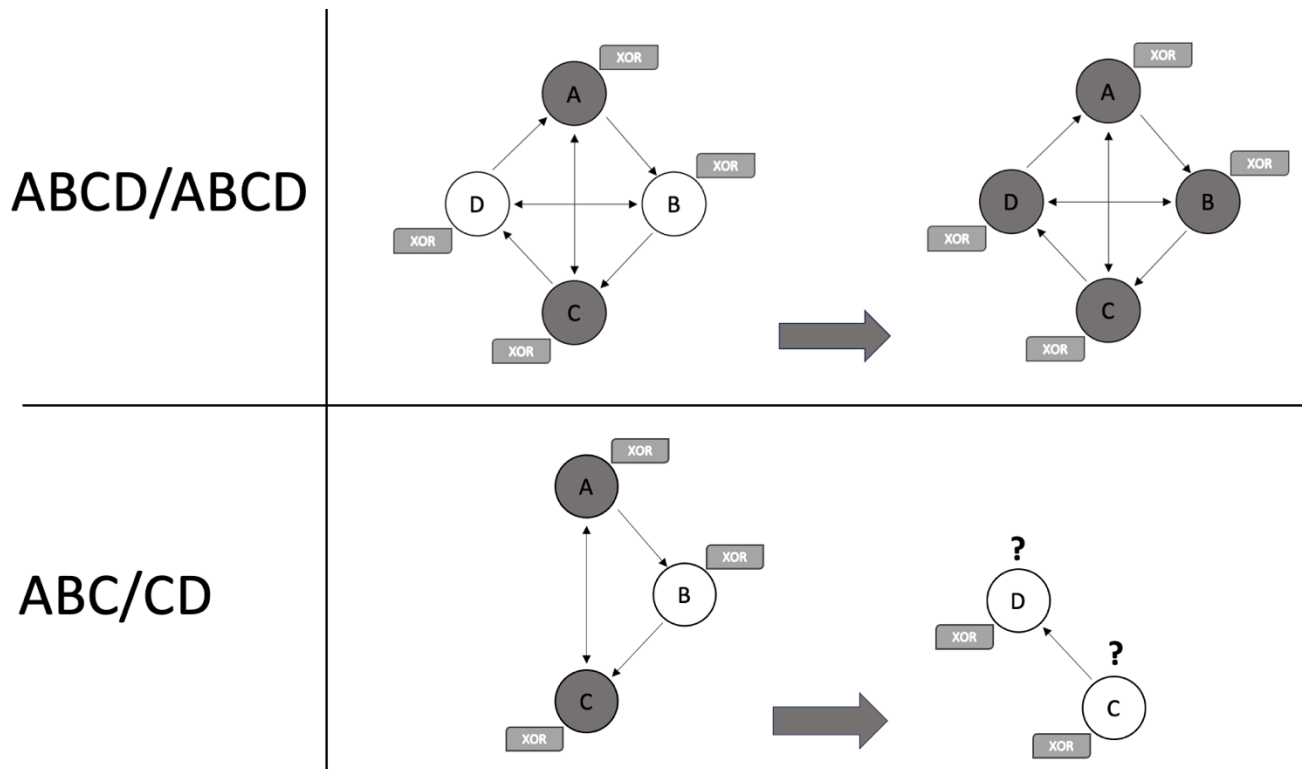




**Fig. 8**

Whereas the state  $ABCD = 1111$  followed with probability 1.0 from the state  $ABCD = 1010$ , we see here that two different states of the system may with equal probability have preceded the state  $ABCD = 0101$ .

One of PyPhi’s primary features is its ability to identify subsets a system that have irreducible causes and effects within it. These subsets are termed “mechanisms” because they are always considered in terms of their causes and effects in relation to other specified subsets of the system, called “purviews” (Albantakis et al. 2023, 10). Thus, with respect to the system from figure 5 PyPhi considers each subset of the powerset of A, B, C, D as a possible mechanism. Each of the subsets “ABCD”, “ABC”, “ABD”, “ACD”, “BCD”, “AB”, “AC”, “AD”, “BC”, “BD”, “CD”, “A”, “B”, “C”, and “D” may be a mechanism with irreducible causes and effects in relation to one of the subset purviews “ABCD”, “ABC”, “ABD”, “ACD”, “BCD”, “AB”, “AC”, “AD”, “BC”, “BD”, “CD”, “A”, “B”, “C”, and “D” (Oizumi et al. 2014, 5-6). One potential mechanism may for instance be the effect of the subset ABC on the subset CD as illustrated in figure 9.

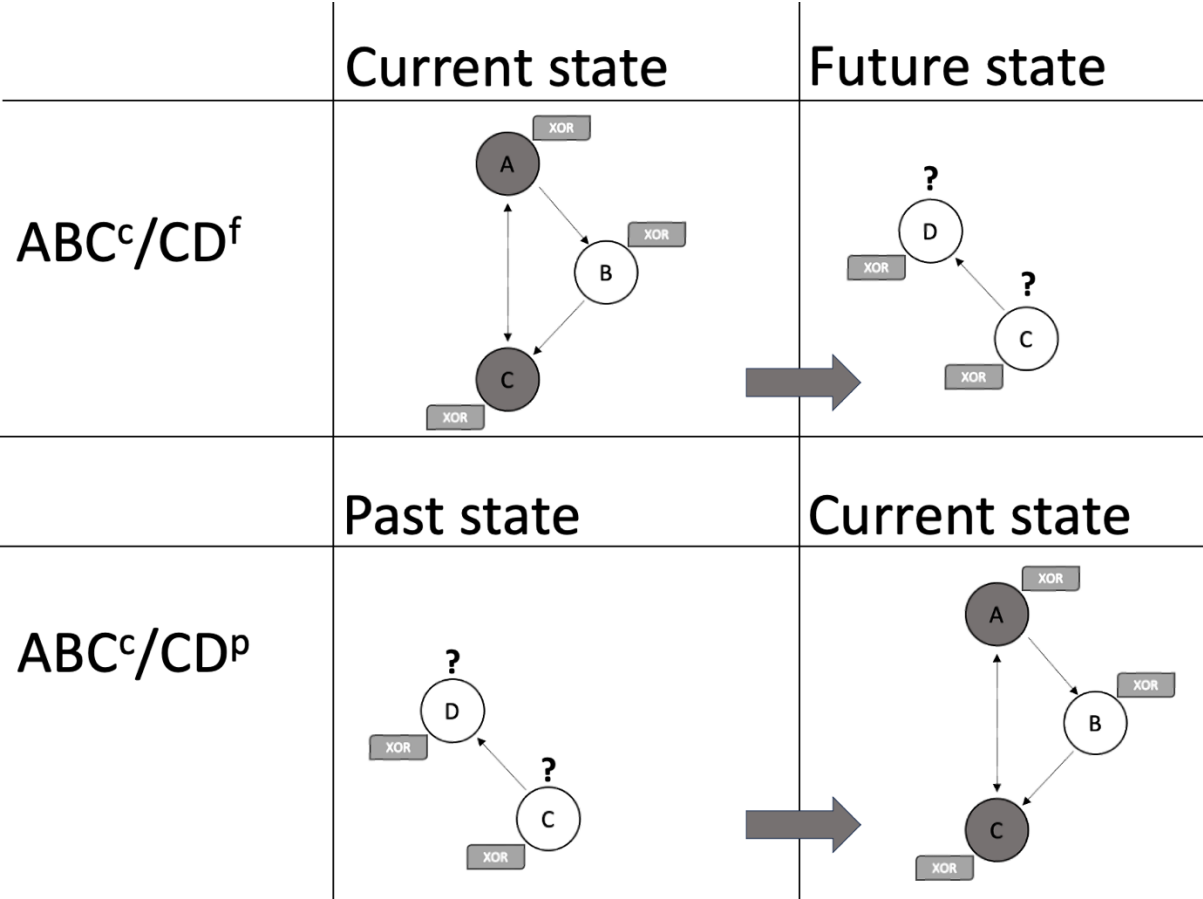


**Fig. 9**

Subset “effect mechanisms” in relation to the subset “purviews”. First, as seen in figure 7 above, when the mechanism ABCD is in the state 1010 at  $t$ , the only possible state at  $t+1$  of the subset purview ABCD is 1111. In similar manner, it may be considered whether the subset “ABC” manifests an effect mechanism with respect to the purview “CD”.

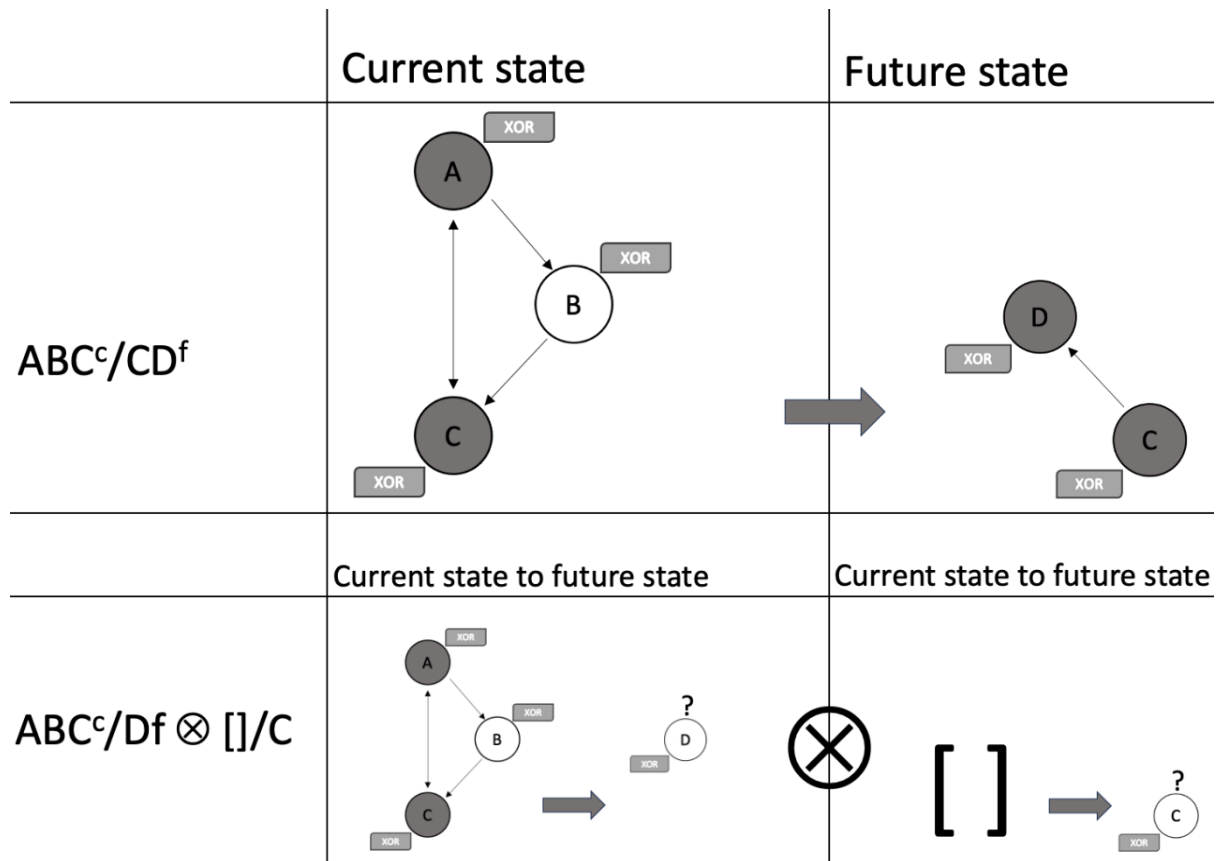
When one subset is considered for its effect mechanism, this is signified by adding a superscript “c” for “current state” to the subset mechanism with potential effects upon the other subset, and a superscript “f” for “future state” to the affected subset:  $ABC^c/CD^f$ .

Likewise, when a subset is considered for its *cause* mechanism in relation to a subset purview, this is signified by adding a superscript “c” to the subset that has potentially been caused by the other subset, and a superscript “p” for “past state” to the subset that has potentially caused the current state:  $ABC^c/CD^p$ , as seen in figure 10 (Oizumi et al. 2014, 6).



**Fig. 10**  
Seen here is first the effect mechanism “ $ABC^c/CD^f$ ”, then the cause mechanism “ $ABC^c/CD^p$ ”. In the case of the effect mechanism  $ABC^c/CD^f$ , the possible effects that the current state of the subset ABC may have on the future state of the purview CD are under consideration. In the case of the cause mechanism  $ABC^c/CD^f$ , the possible states of the past purview CD that may have caused the current state of the subset ABC are under consideration.

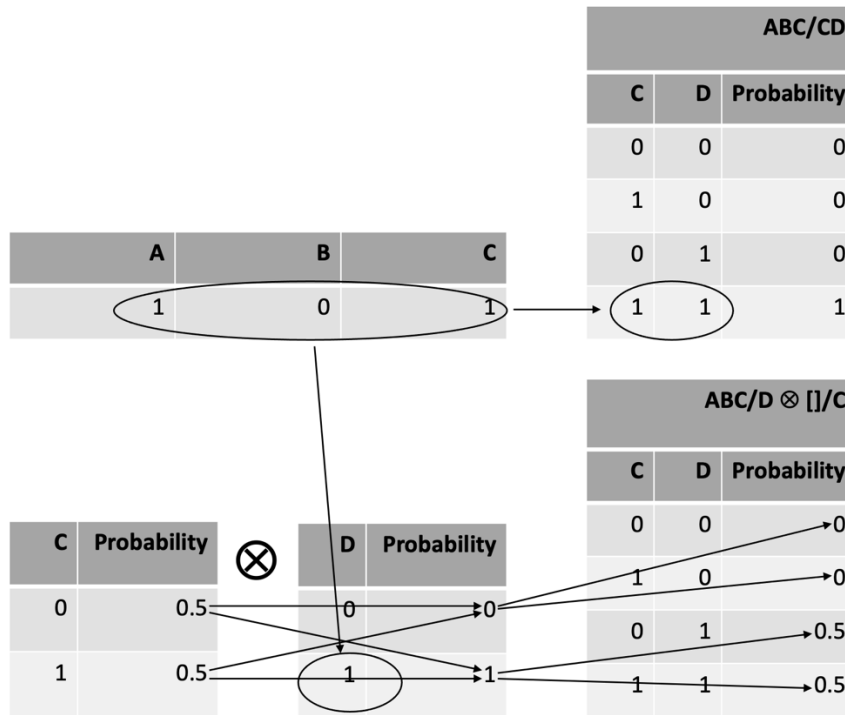
PyPhi evaluates the potential causal irreducibility of a mechanism by comparing the causes and effects that the it has within the system when it is intact to the causes and effects stipulated within the system by the sum of the mechanism considered as two independent subsets (Mayner et al. 2018, 8-11; see also Oizumi et al. 2014, 6-8), as illustrated in figure 11.



**Fig. 11**

The dynamics of the effect mechanism  $ABC^c/CD^f$  is here illustrated first with the mechanism intact, and then with it separated into independent sub-mechanisms, namely how  $ABC$  affects  $D$  and  $C$  in isolation from each other. In order to evaluate the causal irreducibility of the mechanism, PyPhi then compares the information stipulated by the intact mechanism to the information stipulated by the sum of the sub-mechanisms (Mayner et al. 2018 “Supporting Information”, 162-70).

Consider the effect of  $ABC^c$  on the subset purview  $CD^f$ . The only possible state of  $CD^f$  that can follow from  $ABC^c = 101$  is  $CD^f = 11$ . In order to evaluate whether  $ABC^c/CD^f$  can be reduced, PyPhi first splits the single mechanism  $ABC^c/CD^f$  into  $ABC^c/D^f$  and  $[]/C^f$ , then considers them separately before summing them (Mayner et al. 2018, 8-10). As when considered within the intact mechanism  $ABC^c/CD^f$ , the only possible state of  $D^f$  that can follow from  $ABC^c = 101$  is  $D^f = 1$ . However, since  $C^f$  is now considered in isolation from the influence of  $ABC^c$ , it could be in either of the states that are in general possible for it, namely  $C^f = 1$  or  $C^f = 0$ . The tensor product of  $ABC^c/D^f$  and  $[]/C^f$  thus stipulates more possible states than the intact mechanism  $ABC^c/CD^f$  and PyPhi therefore considers  $ABC^c/CD^f$  non-reducible to  $ABC^c/D^f \otimes []/C^f$  (Mayner et al 2018, 8-11: see also Mayner et al. 2018 “Supporting Information”, 162-70). The difference between the two is illustrated in figure 12.



**Fig. 12**

Transition probabilities for first  $ABC^c/CD^f$ , then  $ABC^c/D^f \otimes []/C^f$ . The only possible state of CD that can follow from  $ABC = 101$  is  $CD = 11$ . However, when considered independently, C and D do not specify as much information about the next state of CD, since the tensor product of  $ABC^c/D^f$  and  $[]/C^f$  specify two possible states of CD that could follow from  $ABC = 101$ , namely  $CD = 01$  and  $CD = 11$ .

On the other hand, the tensor product of the causes or effects stipulated by a mechanism's parts may also result in an identical list of possible states. In such cases, IIT considers the subset reducible and thus not a unit mechanism within the system (Mayner et al. 2018, 8-11; see also Mayner et al. 2018 “Supporting Information”, 162-70). For instance, the possible paths along which the mechanism  $AC^c$  could have been caused by the subset  $ABCD^p$  to be in state  $AC^c = 11$  are reducible from  $AC^c/ABCD^p$  to  $A^c/CD^p \otimes C^c/AB^p$ , since both the intact mechanism and the tensor product of the mechanism's parts stipulate four possible states of  $ABCD^p$  previous to  $AC^c = 11$ :  $ABCD^p = 1010$ ,  $ABCD^p = 0110$ ,  $ABCD^p = 1001$ , and  $ABCD^p = 0101$ .

## b. Measuring Integrated Information

“Integrated information” is, when it comes down to it, a measure for the degree to which something is not reducible to its parts. As seen in section 4a, a subset of a system is considered an irreducible subsystem if and only if the subsystem under the overall system's current state is irreducibly caused by the preceding state of at least one other subset (the

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

subsystem’s “cause” mechanism) and irreducibly has effects upon the next state of at least one other subset (the subsystem’s “effect” mechanism). Consider again as an example the system discussed in section 4a. When the whole system is in the state  $ABCD = 1010$ , the current state of the mechanism  $ABC^c$  is irreducibly caused by the subset  $ABCD^p$ , and irreducibly affects the next state of the subset  $CD^f$ . Thus, the subset  $ABC$  is a causally irreducible subsystem within the system when it is in the state  $ABCD = 1010$ .

Specifically, PyPhi considers the integrated information of one mechanism such as  $ABC^c/CD^f$  equal to the distance between the informational content of the intact and partitioned mechanism (Mayner et al. 2018, 8-11; see also Oizumi et al. 2014, 6-8). Since the state of  $CD^f$  is determined with probability 1.0 by the intact mechanism  $ABC^c/CD^f$  but only 0.5 by the sum of the mechanism’s parts  $ABC^c/D^f$  and  $[/math>/ $C^f$ , the integrated information of the mechanism  $ABC^c/CD^f$  is thus 0.5. The integrated information of individual subsystems within a system is represented by lower case “ $\phi$ ” (Oizumi et al. 2014, 4).$

Similar to how the integrated information  $\phi$  for subsystems is found, the integrated information for a whole system, signified by upper case “ $\Phi$ ”, is found by initially separating the network representing the whole system into two independent systems (Mayner et al. 2018, 11-12). PyPhi will then make two lists for comparison. The first list contains all the subsystems with irreducible causes and effects within the intact system, while the second contains the sum  $\phi$  stipulated by the subsystems composing the two independent systems that arise when the intact system is partitioned (Mayner et al. 2018, 12). The integrated information  $\Phi$  for the whole system is equal to the distance between these two lists. This distance is defined as the cost of transforming the  $\phi$  of the subsystems in one list into the  $\phi$  of the subsystems in the other (Oizumi et al. 2014 “Supplementary Methods”, 6-7).

## **5. An Agent-Based Model of the Relationship Between Overfishing and Hypoxia**

This section elaborates on “Hypo\_Socia”, an agent-based model (ABM) designed to examine the suitability of IIT as a general framework for the description and analysis of causal dynamics and emergence within social-ecological systems.

Hypo\_Socia consists of six main agent classes:

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

| Agent Classes        |       |
|----------------------|-------|
| Algae                | (AL)  |
| Zooplankton          | (ZOO) |
| Small Predatory Fish | (SPF) |
| Large Predatory Fish | (LPF) |
| Human Fisheries      | (HF)  |
| Oxygen               | (OX)  |

For each class a set of interaction rules are specified. The full logic behind Hypo\_Socia can be found in the model’s source code in the supplementary material; at present emphasis is put on the rules most relevant for understanding the dynamics of the model, indicated in figure 13 below.

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

| Agents                                        | Reproduction/<br>production         | Consumption            | Disease                   | Rotting                                    |
|-----------------------------------------------|-------------------------------------|------------------------|---------------------------|--------------------------------------------|
| <b>Algae (AL)</b><br>Interval                 | 1 timestep                          | ---                    | ---                       | 13 timesteps                               |
| Duration                                      | ---                                 |                        |                           | 4 timesteps                                |
| Condition                                     | 1 alive AL                          |                        |                           | 1 alive AL                                 |
| Effect                                        |                                     |                        |                           | For each “rotting” turn 1 nearby OX “dies” |
| <b>Zooplankton (ZOO)</b><br>Interval          | 7 timesteps                         | 7-11 timesteps         | ---                       | ---                                        |
| Condition                                     | 1 female + 1 male share cell        | Consume AL else die    | Share cell with other ZOO |                                            |
| Effect                                        | With probability x: 10-20 ZOO added | Consumed AL dies       | With probability x: die   |                                            |
| <b>Small Predatory Fish (SPF)</b><br>Interval | 25 timesteps                        | 14-20 timesteps        | ---                       | ---                                        |
| Condition                                     | 1 female + 1 male share cell        | Consume ZOO else die   | Share cell with other SPF |                                            |
| Effect                                        | With probability x: 5-15 SPF added  | Consumed ZOO dies      | With probability x: die   |                                            |
| <b>Large Predatory Fish (LPF)</b><br>Interval | 20 timesteps                        | 30 timesteps           | ---                       | ---                                        |
| Condition                                     | 1 female + 1 male share cell        | Consume SPF else die   | Share cell with other LPF |                                            |
| Effect                                        | With probability x: 8-12 LPF added  | Consumed SPF dies      | With probability x: die   |                                            |
| <b>Human Fisheries (HF)</b><br>Interval       | ---                                 | 12 timesteps           | ---                       | ---                                        |
| Condition                                     |                                     | Consume LPF else “die” |                           |                                            |
| Effect                                        |                                     | Consumed LPF dies      |                           |                                            |
| <b>Oxygen (OX)</b><br>Interval                | 4 timesteps                         | ---                    | ---                       |                                            |
| Condition                                     | 1 alive AL                          |                        |                           |                                            |

**Fig. 13**

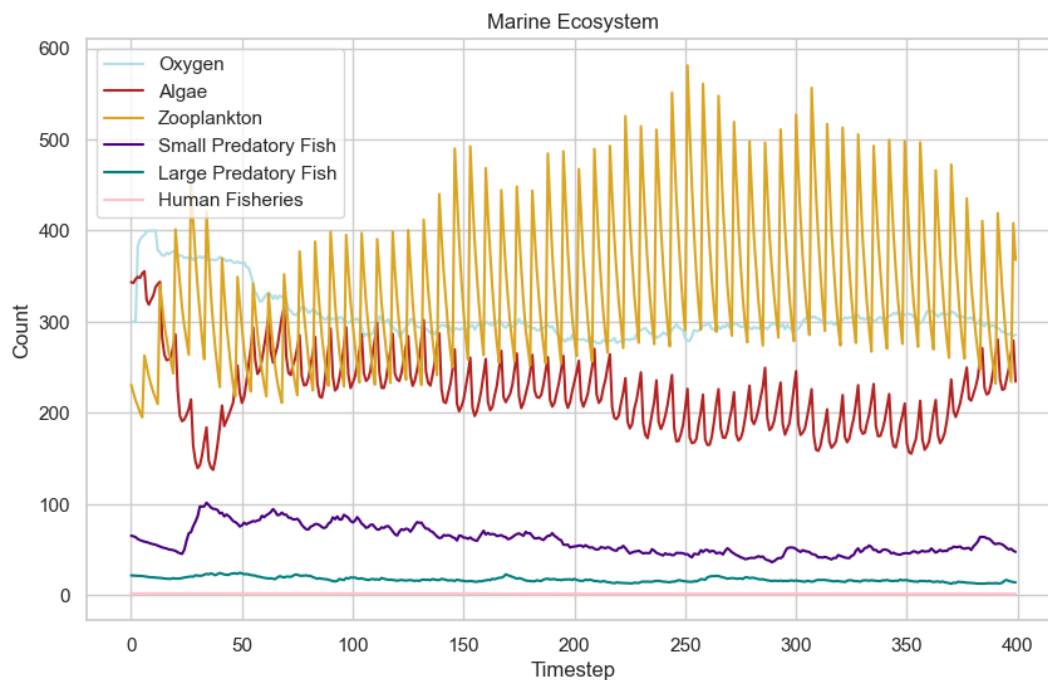
An overview of the central logic behind the behavior of agents in Hypo\_Socia.

During every timestep within the ABM, zooplankton graze on algae, small predatory fish prey on large predatory fish, large predatory fish prey on small predatory fish, and human fisheries catch large predatory fish. If left for too long without being consumed, the algae will begin to rot, depleting oxygen from their surroundings. Several studies have indicated that overfishing of large predatory fish in such an environment may worsen this dynamic by creating a niche for small predatory fish, leading to an increased pressure on the zooplankton populations that dampen outbreaks of algae blooms by regularly grazing on algae (Eriksson et al. 2009; Zaneweld et al. 2016). It was predicted therefore that an ABM simulating these interactions between system elements would display an increased tendency towards hypoxia in reaction to increased manifestations of fisheries.



B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

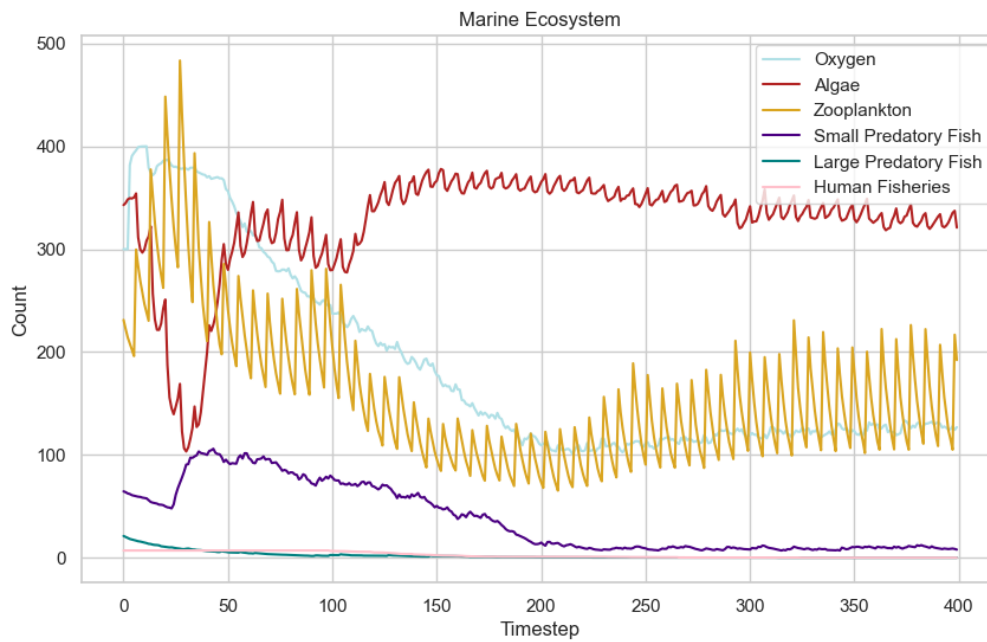
First, initial conditions with relatively few (2) fisheries were stipulated. Figure 14a below is a graph over 400 timesteps averaging over 10 run-throughs of the simulation. As can be seen from this first test-case, an initial rise in the population of small predatory fish correlates to a drop in the zooplankton population, a rise in algae presence, and a drop in dissolved oxygen. The model then stabilizes into a relatively robust pattern.



**Fig. 14a**

Graph describing the number of human fisheries, algae cover, dissolved oxygen, and populations of zooplankton, small predatory fish, and large predatory fish, as expressed by the first run-through of Hypo\_Socia.

When for a second run-through of the model, the number of initial fisheries was increased (7), the pattern of the model differed significantly from the first test-case, as seen in figure 14b. The correlations between the decrease in the population of large predatory fish, increase in the population of small predatory fish, the decrease in the zooplankton population, rise in algae cover, and finally a decrease in dissolved oxygen is amplified.



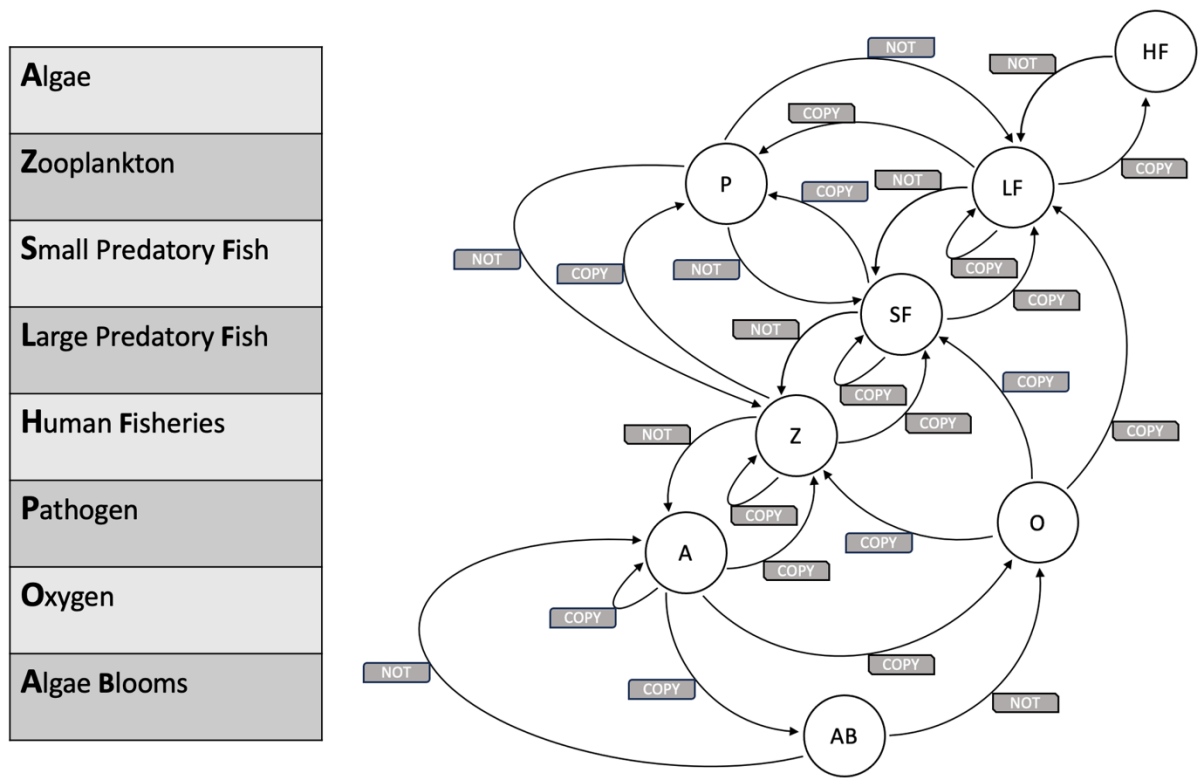
**Fig. 14b**

Graph describing the number of human fisheries, algae cover, dissolved oxygen, and populations of zooplankton, small predatory fish, and large predatory fish, as expressed by the second run-through of Hypo\_Socia. While the effect cannot be linearly ascribed to the increased presence of fisheries—since multiple factors, including the random initial distribution of algae relative to oxygen agents, influence the outcome of the model—some trophic cascade response of increased algae growth relative to the increased presence of fisheries is a plausible interpretation of the data, since no other initial conditions were actively altered.

## 6. Extracting and Preparing ABM Data for Implementation in PyPhi

In this section, it is demonstrated first how data have been extracted from Hypo\_Socia for implementation in PyPhi, then the results of the analysis performed on the ABM using PyPhi are presented.

The first step in preparing data from Hypo\_Socia for implementation within PyPhi was to redescribe the ABM as a Boolean network. This network description should reflect the enabling/restricting relations between the system’s agents, as illustrated in figure 15 below.



**Fig. 15**  
A Boolean network representing the enabling/restricting relations within Hypo\_Socia with “COPY” and “NOT” logic gates. As will be observed, the ABM is represented at the level of all those processes that have causes and effects within the system rather than at the level of individual agents explicitly defined.

Such a network translates in PyPhi into an 8x256 Markov matrix. Data extracted from the model must be normalized into probability values that can be entered into this matrix. The approach selected was to equate the data corresponding to 30 timesteps within Hypo\_Socia with a single transition within the Markov matrix. This level of granularity was chosen because 30 timesteps cover the largest relevant span from cause to effect within the model, namely the time it takes one large predatory fish to die if it does not consume at minimum one small predatory fish. For each node in the network, we then extracted data from timestep 61 to 90, as seen for the node representing small predatory fish in table 1 below.

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

| DATA                                   |                                          |                                            |
|----------------------------------------|------------------------------------------|--------------------------------------------|
| Initial SPF (step 61)                  | SPF offspring (step 61-90)               | Total expected SPF (step 90)               |
| 85.7                                   | 113.2                                    | 198.9                                      |
| SPF disease mortality (step 61-90)     | SPF predation mortality (step 61-90)     | Total expected SPF (step 90)               |
| 65.2                                   | 21                                       | 198.9 - 65.2 - 21 =<br>112.7               |
| SPF resource mortality                 | Result SPF (step 90)                     |                                            |
| 112.7 - 113.2 =<br>32.5                | 80.2                                     |                                            |
| NORMALIZATION                          |                                          |                                            |
| Total SPF (step 61-90)                 | Fraction initial SPF                     | Fraction SPF offspring                     |
| 198.9                                  | 80.2 / 198.9 =<br>0.4032                 | 113.2 / 198.9 =<br>0.5691                  |
| Total fraction SPF disease mortality   | Fraction initial SPF disease mortality   | Fraction SPF offspring disease mortality   |
| 65.2 / 198.9 =<br>0.3278               | 80.2 * 0.3278 =<br>26.29                 | 113 * 0.5 =<br>56.6                        |
|                                        | 26.29 / 198.9 =<br>0.1322                | 56.6 * 0.3278 =<br>18.55                   |
|                                        |                                          | 18.55 / 198.9 =<br>0.0933                  |
| Total fraction SPF predation mortality | Fraction initial SPF predation mortality | Fraction SPF offspring predation mortality |
| 21 / 198.9 =<br>0.1056                 | 80.2 * 0.1056 =<br>8.47                  | 113 * 0.5 =<br>56.6                        |
|                                        | 8.47 / 198.9 =<br>0.0426                 | 56.6 * 0.1056 =<br>5.98                    |
|                                        |                                          | 5.98 / 198.9 =<br>0.0301                   |
| Total fraction SPF resource mortality  | Fraction initial SPF resource mortality  | Fraction SPF offspring resource mortality  |
| 32.2 / 198.9 =<br>0.1619               | 80.2 * 0.1619 =<br>12.98                 | 113 * 0.5 =<br>56.6                        |
|                                        | 12.98 / 198.9 =<br>0.0653                | 56.6 * 0.1619 =<br>9.16                    |
|                                        |                                          | 9.16 / 198.9 =<br>0.0461                   |

**Table 1**

An overview of the mathematical operations used to implement data from Hypo\_Socia in PyPhi. The normalized values represent the probability that the node in question will be enabled or restricted given the interactions it is being considered under. This probability distribution over Hypo\_Socia means, for instance, that if the node “LPP” was active at  $t-1$  then, this restricts the probability of SPF being active at  $t$  to 0.8904. Similar calculations were made for data on all the processes represented by the Boolean network.

## 7. Results

The implementation of data from Hypo\_Socia in PyPhi resulted in a rich analysis of the causal roles played by the subsystems that have irreducible causes and effects within the ABM. The degree to which each subsystem integrates information is illustrated in tables 2 and 3 below. Table 2 is an overview of the values associated with the first run-through “Hypo\_Socia\_1”, which included fewer fisheries (14a). Table 3 shows the values associated with the second run-through “Hypo\_Socia\_2”, which included an increased number of fisheries (14b).

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

| SUBSYS      | $\phi$   | SUBSYS           | $\phi$   | SUBSYS               | $\phi$   | SUBSYS                    | $\phi$   |
|-------------|----------|------------------|----------|----------------------|----------|---------------------------|----------|
| AL          | 0.256325 | AL,OX,AB,ZOO     | 0.002729 | AL,OX,AB,P,HF        | 0.000162 | AL,AB,ZOO,SPF,LPF,P       | 0.017611 |
| OX          | 0.174584 | AL,OX,AB,SPF     | 0.000122 | AL,OX,ZOO,SPF,LPF    | 0.047081 | AL,AB,ZOO,SPF,LPF,HF      | 0.01488  |
| AB          | 0.230775 | AL,OX,AB,P       | 0.000198 | AL,OX,ZOO,SPF,P      | 0.057084 | AL,AB,ZOO,SPF,P,HF        | 0.007834 |
| ZOO         | 0.420762 | AL,OX,ZOO,SPF    | 0.052486 | AL,OX,ZOO,SPF,HF     | 0.030076 | AL,AB,ZOO,LPF,P,HF        | 0.002016 |
| SPF         | 0.085922 | AL,OX,ZOO,LPF    | 0.030484 | AL,OX,ZOO,LPF,P      | 0.05746  | AL,AB,SPF,LPF,P,HF        | 0.00024  |
| LPF         | 0.5      | AL,OX,ZOO,P      | 0.056988 | AL,OX,ZOO,LPF,HF     | 0.030542 | AL,ZOO,SPF,LPF,P,HF       | 0.0302   |
| P           | 0.045034 | AL,OX,SPF,LPF    | 0.063839 | AL,OX,ZOO,P,HF       | 0.0302   | OX,AB,ZOO,SPF,LPF,P       | 8e-06    |
| HF          | 0.017647 | AL,OX,SPF,P      | 0.024813 | AL,OX,SPF,LPF,P      | 0.024813 | OX,AB,ZOO,SPF,LPF,HF      | 8e-06    |
| AL,OX       | 0.04928  | AL,OX,SPF,HF     | 0.019248 | AL,OX,SPF,LPF,HF     | 0.030783 | OX,AB,ZOO,SPF,P,HF        | 8e-06    |
| AL,AB       | 5.8e-05  | AL,OX,LPF,P      | 0.08484  | AL,OX,SPF,P,HF       | 0.024813 | OX,AB,ZOO,LPF,P,HF        | 7e-06    |
| AL,ZOO      | 0.159444 | AL,OX,P,HF       | 0.0302   | AL,OX,LPF,P,HF       | 0.30625  | OX,ZOO,SPF,LPF,P,HF       | 0.054572 |
| AL,SPF      | 0.0396   | AL,AB,ZOO,SPF    | 0.016153 | AL,AB,ZOO,SPF,LPF    | 0.014584 | AB,ZOO,SPF,LPF,P,HF       | 0.004392 |
| AL,P        | 0.033963 | AL,AB,ZOO,LPF    | 0.008003 | AL,AB,ZOO,SPF,P      | 0.017345 | AL,OX,AB,ZOO,SPF,LPF,P    | 0.00585  |
| OX,ZOO      | 0.133456 | AL,AB,ZOO,P      | 0.017377 | AL,AB,ZOO,SPF,HF     | 0.015913 | AL,OX,AB,ZOO,SPF,LPF,HF   | 0.004972 |
| AB,ZOO      | 0.005325 | AL,AB,SPF,LPF    | 0.000151 | AL,AB,ZOO,LPF,P      | 0.017658 | AL,OX,AB,ZOO,SPF,P,HF     | 0.005706 |
| ZOO,SPF     | 0.171809 | AL,AB,SPF,P      | 0.000296 | AL,AB,ZOO,LPF,HF     | 0.007981 | AL,OX,AB,ZOO,LPF,P,HF     | 0.005643 |
| ZOO,LPF     | 0.033706 | AL,AB,SPF,HF     | 0.000175 | AL,AB,ZOO,P,HF       | 0.0151   | AL,OX,AB,SPF,LPF,P,HF     | 0.00015  |
| ZOO,P       | 0.04365  | AL,AB,LPF,P      | 0.000287 | AL,AB,SPF,LPF,P      | 0.000285 | AL,OX,ZOO,SPF,LPF,P,HF    | 0.05587  |
| SPF,LPF     | 0.083375 | AL,AB,P,HF       | 0.00025  | AL,AB,SPF,LPF,HF     | 0.000152 | AL,AB,ZOO,SPF,LPF,P,HF    | 0.011339 |
| SPF,P       | 0.026088 | AL,ZOO,SPF,LPF   | 0.100688 | AL,AB,SPF,P,HF       | 0.000242 | OX,AB,ZOO,SPF,LPF,P,HF    | 8e-06    |
| SPF,HF      | 0.024011 | AL,ZOO,SPF,P     | 0.050463 | AL,AB,LPF,P,HF       | 0.000252 | AL,OX,AB,ZOO,SPF,LPF,P,HF | 0.005752 |
| LPF,P       | 0.024066 | AL,ZOO,SPF,HF    | 0.028344 | AL,ZOO,SPF,LPF,P     | 0.0428   |                           |          |
| LPF,HF      | 0.02205  | AL,ZOO,LPF,P     | 0.024066 | AL,ZOO,SPF,LPF,HF    | 0.0302   |                           |          |
| P,HF        | 0.0151   | AL,ZOO,LPF,HF    | 0.02205  | AL,ZOO,SPF,P,HF      | 0.007834 |                           |          |
| AL,OX,AB    | 3.5e-05  | AL,ZOO,P,HF      | 0.0151   | AL,ZOO,LPF,P,HF      | 0.002016 |                           |          |
| AL,OX,ZOO   | 0.030945 | AL,SPF,LPF,P     | 0.010388 | AL,SPF,LPF,P,HF      | 0.010388 |                           |          |
| AL,OX,SPF   | 0.067066 | AL,SPF,LPF,HF    | 0.029934 | OX,AB,ZOO,SPF,LPF    | 8e-06    |                           |          |
| AL,OX,P     | 0.084516 | AL,SPF,P,HF      | 0.007834 | OX,AB,ZOO,SPF,P      | 8e-06    |                           |          |
| AL,AB,ZOO   | 0.008041 | AL,LPF,P,HF      | 0.002016 | OX,AB,ZOO,SPF,HF     | 8e-06    |                           |          |
| AL,AB,SPF   | 0.000183 | OX,AB,ZOO,SPF    | 8e-06    | OX,AB,ZOO,LPF,P      | 8e-06    |                           |          |
| AL,AB,P     | 0.000301 | OX,AB,ZOO,LPF    | 8e-06    | OX,AB,ZOO,LPF,HF     | 8e-06    |                           |          |
| AL,ZOO,SPF  | 0.100688 | OX,AB,ZOO,P      | 8e-06    | OX,AB,ZOO,P,HF       | 7e-06    |                           |          |
| AL,ZOO,LPF  | 0.033706 | OX,ZOO,SPF,LPF   | 0.136864 | OX,ZOO,SPF,LPF,P     | 0.073544 |                           |          |
| AL,ZOO,P    | 0.072088 | OX,ZOO,SPF,P     | 0.070597 | OX,ZOO,SPF,LPF,HF    | 0.058246 |                           |          |
| AL,SPF,LPF  | 0.0396   | OX,ZOO,SPF,HF    | 0.022527 | OX,ZOO,SPF,P,HF      | 0.023552 |                           |          |
| AL,SPF,P    | 0.010388 | OX,ZOO,LPF,P     | 0.110739 | OX,ZOO,LPF,P,HF      | 0.0441   |                           |          |
| AL,SPF,HF   | 0.018889 | OX,ZOO,LPF,HF    | 0.0441   | AB,ZOO,SPF,LPF,P     | 0.004807 |                           |          |
| AL,LPF,P    | 0.024066 | OX,ZOO,P,HF      | 0.01678  | AB,ZOO,SPF,LPF,HF    | 0.005325 |                           |          |
| AL,P,HF     | 0.0151   | AB,ZOO,SPF,LPF   | 0.005325 | AB,ZOO,SPF,P,HF      | 0.005325 |                           |          |
| OX,AB,ZOO   | 8e-06    | AB,ZOO,SPF,P     | 0.005325 | AB,ZOO,LPF,P,HF      | 0.002016 |                           |          |
| OX,ZOO,SPF  | 0.134797 | AB,ZOO,SPF,HF    | 0.005325 | ZOO,SPF,LPF,P,HF     | 0.0302   |                           |          |
| OX,ZOO,LPF  | 0.138748 | AB,ZOO,LPF,P     | 0.005325 | AL,OX,AB,ZOO,SPF,LPF | 0.004876 |                           |          |
| OX,ZOO,P    | 0.066509 | AB,ZOO,LPF,HF    | 0.005325 | AL,OX,AB,ZOO,SPF,P   | 0.005757 |                           |          |
| AB,ZOO,SPF  | 0.005325 | AB,ZOO,P,HF      | 0.005325 | AL,OX,AB,ZOO,SPF,HF  | 0.005296 |                           |          |
| AB,ZOO,LPF  | 0.005325 | ZOO,SPF,LPF,P    | 0.0428   | AL,OX,AB,ZOO,LPF,P   | 0.00585  |                           |          |
| AB,ZOO,P    | 0.005325 | ZOO,SPF,LPF,HF   | 0.0302   | AL,OX,AB,ZOO,LPF,HF  | 0.002714 |                           |          |
| ZOO,SPF,LPF | 0.124575 | ZOO,SPF,P,HF     | 0.007834 | AL,OX,AB,ZOO,P,HF    | 0.005594 |                           |          |
| ZOO,SPF,P   | 0.071586 | ZOO,LPF,P,HF     | 0.002016 | AL,OX,AB,SPF,LPF,P   | 0.000187 |                           |          |
| ZOO,SPF,HF  | 0.022346 | SPF,LPF,P,HF     | 0.029375 | AL,OX,AB,SPF,LPF,HF  | 0.000103 |                           |          |
| ZOO,LPF,P   | 0.024066 | AL,OX,AB,ZOO,SPF | 0.005377 | AL,OX,AB,SPF,P,HF    | 0.000153 |                           |          |
| ZOO,LPF,HF  | 0.02205  | AL,OX,AB,ZOO,LPF | 0.002721 | AL,OX,AB,LPF,P,HF    | 0.000165 |                           |          |
| ZOO,P,HF    | 0.0151   | AL,OX,AB,ZOO,P   | 0.005753 | AL,OX,ZOO,SPF,LPF,P  | 0.056904 |                           |          |
| SPF,LPF,P   | 0.021225 | AL,OX,AB,SPF,LPF | 0.000102 | AL,OX,ZOO,SPF,LPF,HF | 0.048227 |                           |          |
| SPF,LPF,HF  | 0.0302   | AL,OX,AB,SPF,P   | 0.000196 | AL,OX,ZOO,SPF,P,HF   | 0.044698 |                           |          |

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

|          |          |                 |          |                    |          |  |
|----------|----------|-----------------|----------|--------------------|----------|--|
| SPF,P,HF | 0.007834 | AL,OX,AB,SPF,HF | 0.000115 | AL,OX,ZOO,LPF,P,HF | 0.0441   |  |
| LPF,P,HF | 0.002016 | AL,OX,AB,LPF,P  | 0.000194 | AL,OX,SPF,LPF,P,HF | 0.024813 |  |

**Table 2**

A list of all subsystems that have irreducible causes and effects within Hypo\_Socia\_1 and the associated  $\phi$  values that quantify the degree to which each subsystem has irreducible causes and effects within Hypo\_Socia\_1.

| SUBSYS      | $\phi$   | SUBSYS           | $\phi$   | SUBSYS               | $\phi$   | SUBSYS                    | $\phi$   |
|-------------|----------|------------------|----------|----------------------|----------|---------------------------|----------|
| AL          | 0.26175  | AL,OX,AB,ZOO     | 0.002673 | AL,OX,AB,P,HF        | 4.6e-05  | AL,AB,ZOO,SPF,LPF,P       | 0.010277 |
| OX          | 0.141559 | AL,OX,AB,SPF     | 6.3e-05  | AL,OX,ZOO,SPF,LPF    | 0.060563 | AL,AB,ZOO,SPF,LPF,HF      | 0.010416 |
| AB          | 0.2509   | AL,OX,AB,P       | 4.3e-05  | AL,OX,ZOO,SPF,P      | 0.064529 | AL,AB,ZOO,SPF,P,HF        | 0.008476 |
| ZOO         | 0.473512 | AL,OX,ZOO,SPF    | 0.074744 | AL,OX,ZOO,SPF,HF     | 0.004401 | AL,AB,ZOO,LPF,P,HF        | 0.007819 |
| SPF         | 0.109522 | AL,OX,ZOO,LPF    | 0.04664  | AL,OX,ZOO,LPF,P      | 0.051025 | AL,AB,SPF,LPF,P,HF        | 5.2e-05  |
| LPF         | 0.5      | AL,OX,ZOO,P      | 0.066153 | AL,OX,ZOO,LPF,HF     | 0.038638 | AL,ZOO,SPF,LPF,P,HF       | 0.003244 |
| P           | 0.044063 | AL,OX,SPF,LPF    | 0.050863 | AL,OX,ZOO,P,HF       | 0.013118 | OX,AB,ZOO,SPF,LPF,P       | 3.3e-05  |
| HF          | 0.055566 | AL,OX,SPF,P      | 0.0292   | AL,OX,SPF,LPF,P      | 0.026488 | OX,AB,ZOO,SPF,LPF,HF      | 3.3e-05  |
| AL,OX       | 0.074209 | AL,OX,SPF,HF     | 0.003999 | AL,OX,SPF,LPF,HF     | 0.019275 | OX,AB,ZOO,SPF,P,HF        | 3.2e-05  |
| AL,AB       | 2e-05    | AL,OX,LPF,P      | 0.038975 | AL,OX,SPF,P,HF       | 0.019955 | OX,AB,ZOO,LPF,P,HF        | 3.2e-05  |
| AL,ZOO      | 0.170663 | AL,OX,P,HF       | 0.018609 | AL,OX,LPF,P,HF       | 0.005963 | OX,ZOO,SPF,LPF,P,HF       | 0.019594 |
| AL,SPF      | 0.032894 | AL,AB,ZOO,SPF    | 0.019951 | AL,AB,ZOO,SPF,LPF    | 0.017558 | AB,ZOO,SPF,LPF,P,HF       | 0.002484 |
| AL,P        | 0.040188 | AL,AB,ZOO,LPF    | 0.008238 | AL,AB,ZOO,SPF,P      | 0.007885 | AL,OX,AB,ZOO,SPF,LPF,P    | 0.005001 |
| OX,ZOO      | 0.150106 | AL,AB,ZOO,P      | 0.015267 | AL,AB,ZOO,SPF,HF     | 0.004474 | AL,OX,AB,ZOO,SPF,LPF,HF   | 0.004293 |
| AB,ZOO      | 0.010925 | AL,AB,SPF,LPF    | 4.3e-05  | AL,AB,ZOO,LPF,P      | 0.007819 | AL,OX,AB,ZOO,SPF,P,HF     | 0.004964 |
| ZOO,SPF     | 0.097513 | AL,AB,SPF,P      | 5.1e-05  | AL,AB,ZOO,LPF,HF     | 0.008238 | AL,OX,AB,ZOO,LPF,P,HF     | 0.005019 |
| ZOO,LPF     | 0.008238 | AL,AB,SPF,HF     | 4.7e-05  | AL,AB,ZOO,P,HF       | 0.014267 | AL,OX,AB,SPF,LPF,P,HF     | 3.4e-05  |
| ZOO,P       | 0.039938 | AL,AB,LPF,P      | 6.1e-05  | AL,AB,SPF,LPF,P      | 5e-05    | AL,OX,ZOO,SPF,LPF,P,HF    | 0.019594 |
| SPF,LPF     | 0.026488 | AL,AB,P,HF       | 5.8e-05  | AL,AB,SPF,LPF,HF     | 4.3e-05  | AL,AB,ZOO,SPF,LPF,P,HF    | 0.003244 |
| SPF,P       | 0.054119 | AL,ZOO,SPF,LPF   | 0.033231 | AL,AB,SPF,P,HF       | 5.3e-05  | OX,AB,ZOO,SPF,LPF,P,HF    | 3.3e-05  |
| SPF,HF      | 0.004666 | AL,ZOO,SPF,P     | 0.063881 | AL,AB,LPF,P,HF       | 5.8e-05  | AL,OX,AB,ZOO,SPF,LPF,P,HF | 0.00499  |
| LPF,P       | 0.002281 | AL,ZOO,SPF,HF    | 0.004498 | AL,ZOO,SPF,LPF,P     | 0.020638 |                           |          |
| LPF,HF      | 0.013294 | AL,ZOO,LPF,P     | 0.007819 | AL,ZOO,SPF,LPF,HF    | 0.010416 |                           |          |
| P,HF        | 0.006104 | AL,ZOO,LPF,HF    | 0.008238 | AL,ZOO,SPF,P,HF      | 0.008657 |                           |          |
| AL,OX,AB    | 3.9e-05  | AL,ZOO,P,HF      | 0.014287 | AL,ZOO,LPF,P,HF      | 0.007819 |                           |          |
| AL,OX,ZOO   | 0.046599 | AL,SPF,LPF,P     | 0.007294 | AL,SPF,LPF,P,HF      | 0.003244 |                           |          |
| AL,OX,SPF   | 0.102884 | AL,SPF,LPF,HF    | 0.010416 | OX,AB,ZOO,SPF,LPF    | 3.3e-05  |                           |          |
| AL,OX,P     | 0.102616 | AL,SPF,P,HF      | 0.007294 | OX,AB,ZOO,SPF,P      | 3.3e-05  |                           |          |
| AL,AB,ZOO   | 0.010579 | AL,LPF,P,HF      | 0.002981 | OX,AB,ZOO,SPF,HF     | 3.3e-05  |                           |          |
| AL,AB,SPF   | 5.2e-05  | OX,AB,ZOO,SPF    | 3.3e-05  | OX,AB,ZOO,LPF,P      | 3.3e-05  |                           |          |
| AL,AB,P     | 6.00E-05 | OX,AB,ZOO,LPF    | 3.4e-05  | OX,AB,ZOO,LPF,HF     | 3.3e-05  |                           |          |
| AL,ZOO,SPF  | 0.091675 | OX,AB,ZOO,P      | 3.2e-05  | OX,AB,ZOO,P,HF       | 3.2e-05  |                           |          |
| AL,ZOO,LPF  | 0.008238 | OX,ZOO,SPF,LPF   | 0.060563 | OX,ZOO,SPF,LPF,P     | 0.035375 |                           |          |
| AL,ZOO,P    | 0.0641   | OX,ZOO,SPF,P     | 0.065637 | OX,ZOO,SPF,LPF,HF    | 0.023075 |                           |          |
| AL,SPF,LPF  | 0.026488 | OX,ZOO,SPF,HF    | 0.003575 | OX,ZOO,SPF,P,HF      | 0.004443 |                           |          |
| AL,SPF,P    | 0.007294 | OX,ZOO,LPF,P     | 0.035703 | OX,ZOO,LPF,P,HF      | 0.01345  |                           |          |
| AL,SPF,HF   | 0.00393  | OX,ZOO,LPF,HF    | 0.038638 | AB,ZOO,SPF,LPF,P     | 0.002578 |                           |          |
| AL,LPF,P    | 0.002281 | OX,ZOO,P,HF      | 0.002166 | AB,ZOO,SPF,LPF,HF    | 0.010416 |                           |          |
| AL,P,HF     | 0.024519 | AB,ZOO,SPF,LPF   | 0.010925 | AB,ZOO,SPF,P,HF      | 0.002636 |                           |          |
| OX,AB,ZOO   | 3.3e-05  | AB,ZOO,SPF,P     | 0.002654 | AB,ZOO,LPF,P,HF      | 0.00447  |                           |          |
| OX,ZOO,SPF  | 0.147833 | AB,ZOO,SPF,HF    | 0.003566 | ZOO,SPF,LPF,P,HF     | 0.003244 |                           |          |
| OX,ZOO,LPF  | 0.066119 | AB,ZOO,LPF,P     | 0.00481  | AL,OX,AB,ZOO,SPF,LPF | 0.004295 |                           |          |
| OX,ZOO,P    | 0.044999 | AB,ZOO,LPF,HF    | 0.008238 | AL,OX,AB,ZOO,SPF,P   | 0.004964 |                           |          |
| AB,ZOO,SPF  | 0.010925 | AB,ZOO,P,HF      | 0.002168 | AL,OX,AB,ZOO,SPF,HF  | 0.004399 |                           |          |
| AB,ZOO,LPF  | 0.008238 | ZOO,SPF,LPF,P    | 0.020638 | AL,OX,AB,ZOO,LPF,P   | 0.00503  |                           |          |
| AB,ZOO,P    | 0.003256 | ZOO,SPF,LPF,HF   | 0.010416 | AL,OX,AB,ZOO,LPF,HF  | 0.002689 |                           |          |
| ZOO,SPF,LPF | 0.033231 | ZOO,SPF,P,HF     | 0.004419 | AL,OX,AB,ZOO,P,HF    | 0.004994 |                           |          |
| ZOO,SPF,P   | 0.043331 | ZOO,LPF,P,HF     | 0.007819 | AL,OX,AB,SPF,LPF,P   | 2.8e-05  |                           |          |
| ZOO,SPF,HF  | 0.00358  | SPF,LPF,P,HF     | 0.003244 | AL,OX,AB,SPF,LPF,HF  | 5.4e-05  |                           |          |
| ZOO,LPF,P   | 0.007819 | AL,OX,AB,ZOO,SPF | 0.004853 | AL,OX,AB,SPF,P,HF    | 3e-05    |                           |          |
| ZOO,LPF,HF  | 0.008238 | AL,OX,AB,ZOO,LPF | 0.002698 | AL,OX,AB,LPF,P,HF    | 4.8e-05  |                           |          |
| ZOO,P,HF    | 0.002109 | AL,OX,AB,ZOO,P   | 0.00502  | AL,OX,ZOO,SPF,LPF,P  | 0.035375 |                           |          |
| SPF,LPF,P   | 0.015338 | AL,OX,AB,SPF,LPF | 5.5e-05  | AL,OX,ZOO,SPF,LPF,HF | 0.023075 |                           |          |

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

|            |          |                 |         |                    |          |  |
|------------|----------|-----------------|---------|--------------------|----------|--|
| SPF,LPF,HF | 0.010416 | AL,OX,AB,SPF,P  | 2.9e-05 | AL,OX,ZOO,SPF,P,HF | 0.008384 |  |
| SPF,P,HF   | 0.006415 | AL,OX,AB,SPF,HF | 6.3e-05 | AL,OX,ZOO,LPF,P,HF | 0.01345  |  |
| LPF,P,HF   | 0.002981 | AL,OX,AB,LPF,P  | 4.6e-05 | AL,OX,SPF,LPF,P,HF | 0.015794 |  |

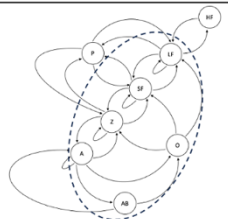
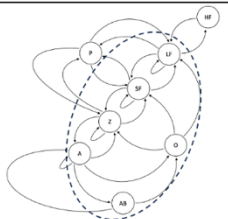
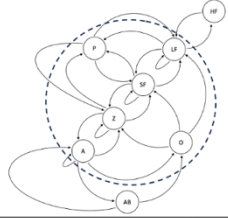
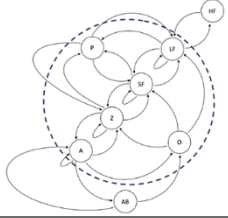
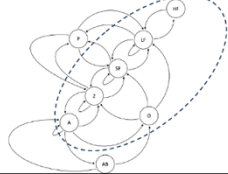
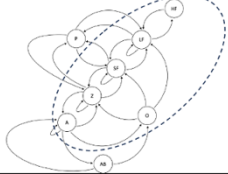
**Table 3**

A list of all subsystems that have irreducible causes and effects within Hypo\_Socia\_1 and the associated  $\phi$  values that quantify the degree to which each subsystem has irreducible causes and effects within Hypo\_Socia\_2.

As can be seen by comparing the tables, human fisheries (HF), small predatory fish (SPF), algae (AL) and algae blooms (AB) all play more causally significant roles in Hypo\_Socia\_2 than in Hypo\_Socia\_1 from timestep 61 to 90. The differences in causal efficacy within the system are quantified as the  $\phi$  values 0.055566 (HF), 0.109522 (SPF), 0.26175 (AL) and 0.2509 (AB) within Hypo\_Socia\_2 against 0.017647 (HF), 0.085922 (SPF), 0.256325 (AL) and 0.230775 (AB) within Hypo\_Socia\_1. This links well with both our expectations and the actual behavior of the ABM. The amplified  $\phi$  of SPF in Hypo\_Socia\_2 relative to its  $\phi$  in Hypo\_Socia\_1 quantifies the causal niche created in the system for SPF when the population of large predatory fish (LPF) is rapidly depleted by human fisheries (HF). Likewise, the expected and observed dynamics of the trophic cascade are quantitatively reflected in the increased causal influence afforded to both AL and AB in this scenario.

Other conceptually fruitful comparisons can be made, but for now just one more is singled out. Note that the subsystem consisting of SPF and LPF in interaction has a significantly greater causal niche within Hypo\_Socia\_1 than it has within Hypo\_Socia\_2, with a  $\phi$  value of 0.083375 against 0.026488. This difference quantifies the degree of causal constraint that LPF should exert on SPF from timestep 61 to 90 in order for the system to stabilize into the pattern observed in Hypo\_Socia\_1 as opposed to the one observed in Hypo\_Socia\_2.

Furthermore, the integrated information  $\Phi$  across the whole ABM from timestep 61-90 was considered. This value is of interest since a system that exhibits global integrated information will also be likely to exhibit behavior at the system level that cannot be gleaned simply from the behavior of its components.  $\Phi$ , in short, quantifies the degree to which we should expect a system to behave unpredictably relative to the behavior of its parts. Since the calculation of  $\Phi$  for a stochastic system such as Hypo\_Socia is highly computationally demanding, the approach has been to perform cross-sectional calculations over the whole system, as illustrated in figure 16.

| Hypo_Socia_1 $\Phi$                                                               |                                             | Hypo_Socia_2 $\Phi$                                                                |                                             |
|-----------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
|  | A, OX, AB, ZOO, SF, LF<br>$\Phi$ : 0.344566 |  | A, OX, AB, ZOO, SF, LF<br>$\Phi$ : 0.182369 |
|  | A, OX, ZOO, SF, LF, P<br>$\Phi$ : 0.909634  |  | A, OX, ZOO, SF, LF, P<br>$\Phi$ : 0.609987  |
|  | A, OX, ZOO, SF, LF, HF<br>$\Phi$ : 0.429129 |  | A, OX, ZOO, SF, LF, HF<br>$\Phi$ : 0.301488 |

**Fig. 16**

Illustration of the cross-sections across which the integrated information  $\Phi$  for the whole systems Hypo\_Socia\_1 and Hypo\_Socia\_2 was calculated. Though this approach does not provide an exact measure of the  $\Phi$  associated with the ABM, it does offer a less computationally demanding way to evaluate whether it is likely to have non-zero  $\Phi$ .

These calculations confirm that Hypo\_Socia exhibits  $\Phi$  across the sections “A, OX, AB, ZOO, SF, LF”, “A, OX, ZOO, SF, LF, P” and “A, OX, ZOO, SF, LF, HF”. Thus, information is likely integrated across the whole system, and it should therefore be expected to exhibit behavior that is unpredictable relative to the behavior of its components.

Note, finally, that the  $\Phi$  associated with each cross-section of the system is lower in Hypo\_Socia\_2 relative to Hypo\_Socia\_1. This difference illustrates well the way in which PyPhi can be utilized to quantify the degree to which the activities of specific components affect the overall cohesion and resilience of a system, since it may reasonably be assumed that the greater  $\Phi$  generated by Hypo\_Socia\_1 relative to Hypo\_Socia\_2 is connected to the less-perturbed interactions associated with LPF found within it.

**7. Discussion**

The findings in this study emphasize the multi-layered nature of causal relationships in social-ecological systems. While it has generally proven difficult to forecast the behavior of such highly complex systems strictly by observing the behavior of their parts, the notion that novel physical characteristics may emerge at hierarchically differentiated levels of physical



B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

systems remains to some extent controversial. Presumably the stalemate between reductionism and the unpredictable behavior of complex systems must eventually end either with the formulation of complete bridge laws between different hierarchical levels in reductive pathways or a conception of emergence that can be shown to harmonize well with the naturalistic attitude of science. The causal account of emergence offered by IIT is promising in this regard because it is premised not on mereological speculations but rather the testable causal constraints exerted among system components.

The highly varied groupings of Hypo\_Socia’s specified network nodes as subsystems with non-zero  $\phi$  returned by the integrated information analysis suggest that this type of system manifests not just at the level of intuitively clear-cut units, such as individual organisms, but also in the interactions between individuals, and at the wider level of such interactive processes cross-sectionally constraining one another. Thus, when the  $\phi$ ’s of the algae (AL: 0.256325), oxygen (OX: 0.174584), algae blooms (AB: 0.230775) and zooplankton (ZOO: 0.420762) subsystems do not sum to the  $\phi$  of the group (AL, OX, AB, ZOO: 0.002673) this is because the interactions between AL, OX, AB and ZOO constitute a unique subsystem over and above the distinct components interacting in the group. The  $\phi$  value 0.002673 quantifies the degree to which the set of interactions between AL, OX, AB and ZOO exerts and is subject to distinct causal pressures within Hypo\_Socia that are fundamentally different from any of those associated with either AL, OX, AB, ZOO or any lower-level combination of these in isolation.

This insight has not only conceptual, but also practical consequences. For instance, while the interactions between AL, OX, AB and ZOO are causally irreducible within Hypo\_Socia to the degree  $\phi$  0.002673, the causal irreducibility of the smaller subset consisting of AL, OX and ZOO is markedly higher at  $\phi$  0.030945. Interestingly, the AB node serves to a large extent as a feed-forward mechanism for the AL node within the Boolean network, effectuating relationships that it would be unfeasible to attach directly to AL. One example of how this works is that, since AL is supposed to enable the OX node under some circumstances but restrict it under others, AB takes over the role of restricting OX. The distinction between the causal profiles of “AL, OX, AB, ZOO” and “AL, OX, ZOO” thus quantitatively indicates where in the system interventional resources should be focused by exposing that interventions directly aimed at AB would likely influence the overall system to a lesser degree than interventions aimed at AL.

The utilization of PyPhi suggested in this study has, I believe, the potential to open new avenues within environmental modelling. However, the study’s reliance on a limited

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

number of simulations implicates a need for future studies with more robust statistical foundations. Since the data extracted from Hypo\_Socia\_1 and Hypo\_Socia\_2 was generated by averaging over 10 run-throughs for each scenario, other factors, such as the initial distribution of agents, could be contributing to the pattern arising in Hypo\_Socia\_2 that has here been interpreted as trophic cascade. Additionally, although targeted at social-ecological systems precisely with the multifaceted causal relationships that characterize these in mind, the social dynamics in Hypo\_Socia are limited to the competition for resources between different fisheries. Future research should accentuate feedback mechanisms between ecological and social dynamics, for example by explicitly introducing consumers depending on the fisheries for food.

## 8. Conclusion

By first designing the ABM “Hypo\_Socia” and then employing the Python software package PyPhi to analyze it, this study has demonstrated the potential of IIT’s formal framework to provide novel insights into the causal dynamics of social-ecological systems. The integrated information analysis of Hypo\_Socia has highlighted that complex, layered interplays of causal relationships define such systems, accentuating the necessity of scrutinizing them at all possible levels of organization. The key findings of the study demonstrate the potential of this approach to contribute with quantitative insights to the development of ecosystem management and intervention strategies.

In view of the exploratory nature of the approach, further study is needed to gauge its true potential. In particular the need for implementing in PyPhi data extracted from models that simulate the complex feedback mechanisms between ecological and social dynamics at higher resolutions has been emphasized. Taken in this direction, the application of PyPhi in the context of social-ecological systems could prove a fruitful tool for improving our knowledge of the dynamic relationship between humans and ecosystems.

## References

- Albantakis, L., Barbosa, L., Findlay, G., Grasso, M., Haun, A. M., Marshall, W., Mayner, W. G. P., Zaeemzadeh, A., Boly, M., Juel, B. E., Sasai, S., Fujii, K., David, I., Hendren, J., Lang, J. P., & Tononi, G. (2023). “Integrated information theory 4.0: Formulating the properties of phenomenal existence in physical terms”. *PLOS Computational Biology*, 19(2), e1011465. <https://doi.org/10.1371/journal.pcbi.1011465>
- Albantakis, L., Marshall, W., Hoel, E., & Tononi, G. (2019). “What Caused What? A

B. Christensen (2025), “Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi”. Research Report protected by copyright laws.

Quantitative Account of Actual Causation Using Dynamical Causal Networks”.

*Entropy*, 21(5), 459. <https://doi.org/10.3390/e21050459>

Banitz, T., Hertz, T., Johansson, L., Lindkvist, E., Martínez-Peña, R., Radosavljevic, S., Schlüter, M., Wennberg, K., Ylikoski, P., Grimm, V. “Visualization of causation in social-ecological systems”. *Ecology and Society*, 27(1): 31  
<https://doi.org/10.5751/ES-13030-270131>

Engel, D., & Malone, T. W. (2018). Integrated information as a metric for group interaction. *PLOS ONE*, 13(10), e0205335. <https://doi.org/10.1371/journal.pone.0205335>

Eriksson, B. K., Sieben, K., Eklöf, J., Ljunggren, L., Olsson, J., Casini, M., & Bergström, U. (2011). Effects of altered offshore food webs on coastal ecosystems emphasize the need for cross-ecosystem management. *Ambio*, 40(7), 786-797.  
<https://doi.org/10.1007/s13280-011-0185-6>

Fischer, D., Mostaghim, S., & Albantakis, L. (2020). How cognitive and environmental constraints influence the reliability of simulated animats in groups. *PLOS ONE*, 15(3), e0228879. <https://doi.org/10.1371/journal.pone.0228879>

Marshall, W., Albantakis, L., & Tononi, G. (2018). “Black-boxing and cause-effect power”. *PLOS Computational Biology*, 14(4), <https://doi.org/10.1371/journal.pcbi.1006114>

Marshall, W., Kim, H., Walker, S. I., Tononi, G., & Albantakis, L. (2017). How causal analysis can reveal autonomy in models of biological systems. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 375(2109), 20160358. <https://doi.org/10.1098/rsta.2016.0358>

Mayner, W. G. P., Marshall, W., Albantakis, L., Findlay, G., Marchman, R., & Tononi, G. (2018). PyPhi: A toolbox for integrated information theory. *PLoS Computational Biology*, 14(7), e1006343. <https://doi.org/10.1371/journal.pcbi.1006343>

Mediano, P. A. M., Seth, A. K., Barrett, A. B., Carhart-Harris, R. L., Bor, D., & Rosas, F. E. (2022). Integrated information as a common signature of dynamical and information processing complexity. *Nature Communications*, 13(1), 1132.  
<https://doi.org/10.1038/s41467-022-28663-y>

Niizato, T., Murakami, H., Gunji, Y. P., & Ito, K. (2020). Finding continuity and discontinuity in fish schools via integrated information theory. *PLOS ONE*, 15(2), e0229573. <https://doi.org/10.1371/journal.pone.0229573>

Oizumi, M., Albantakis, L., & Tononi, G. (2014). From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0. *PLoS Computational Biology*, 10(5), e1003588.

B. Christensen (2025), "Assessing Causal Dynamics and Levels of Emergence in Social-Ecological Systems with PyPhi". Research Report protected by copyright laws.

<https://doi.org/10.1371/journal.pcbi.1003588>

Schlüter, M., Brelsford, C., Ferraro, P. J., Orach, K., Qiu, M., & Smith, M. D. (2023).

"Unraveling complex causal processes that affect sustainability requires more integration between empirical and modeling approaches". *PNAS*, 120(41).

<https://doi.org/10.1073/pnas.2215676120>

Tononi, G. (2004). An Information Integration Theory of Consciousness. *BMC*

*Neuroscience*, 5, 42. <https://doi.org/10.1186/1471-2202-5-42>

Tononi, G., Sporns, O., & Edelman, G. M. (1996). A complexity measure for selective matching of signals by the brain. *Proc Natl Acad Sci U S A*, 93(8), 3422-3427.

<https://doi.org/10.1073/pnas.93.8.3422>

Zaneveld, J. R., Burkepile, D. E., Shantz, A. A., Pritchard, C. E., McMinds, R., Payet, J. P., Welsh, R., Correa, A. M. S., Lemoine, N. P., Rosales, S., Fuchs, C., & Vega Thurber, R. L. (2016). Overfishing and nutrient pollution interact with temperature to disrupt coral reefs down to microbial scales. *Nature Communications*, 7, 11833.

<https://doi.org/10.1038/ncomms11833>